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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	MK GreenCode
Selected Domain	Domain 1: The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth
Project / Solution Name	Workforce Futures
Name (Leader)	Wong Yik Sheng
Email & Phone (Leader)	yswong2302@gmail.com & +60 11-11682302
University Affiliation(s)	Universiti Malaya
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

AI is changing which skills employers ask for faster than workers, students and career advisers can see. With the existence of AI, redundant jobs will be greatly replaced and skills related to AI operation in jobs is upscaling too. In Malaysia, more than 600,000 workers are expected to need reskilling within three to five years .

The people most affected are mid-career workers in routine office roles, fresh graduates choosing a first job, TVET students, and the HR managers and career centres who plan their training. Take Mei Ling, an office clerk for 18 years, she keys in orders and prepares invoices in Excel. However, public job-post data shows exactly these skills cooling. The share of posts asking for spreadsheets and data handling fell a lot since generative AI which can easily handle this job exist, while databases and SQL rose a lot to handle AI memory.

Today she cannot see which of her skills are fading, which future role fits her, or what to learn first. Advice is generic “learn AI”, courses are scattered and stated with unclear prices, and her manager plans the team’s training in a spreadsheet. The result is wasted training money, people reskilling into the wrong roles, and workers left behind, an economic and social cost that falls hardest on those with the least time and money to experiment.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Workforce Futures is a deployed, AI-assisted career intelligence platform that maps an individual's current skills to emerging roles and generates a costed, step-by-step reskilling pathway. It pairs a deterministic scoring engine with a large language model (LLM) operating under strict guardrails, so every recommendation is explainable and auditable.

(1) Profile ingestion: users upload a CV (PDF/DOCX), complete a structured form or enter free text in English, Bahasa Malaysia or Chinese. Personal identifiers are redacted before the LLM extracts skills against a fixed taxonomy of 39 skill clusters, citing the user's own words as evidence for each skill.
(2) Labour-market signal layer: a scheduled job ingests public job postings (Hacker News "Who is hiring?") and AI research metadata (arXiv) daily, computes period-over-period demand shifts per skill cluster and stores dated snapshots.
(3) Career matching: a transparent weighted model ranks 20 future-facing careers mapped to ESCO occupations, combining skill coverage, demand outlook, transferable skills and interest fit, adjusted for occupational GenAI exposure (ILO index). Each score exposes its components, skills gap and three demand scenarios.
(4) Pathway generation: a constraint-aware planner sequences roles from entry to senior level and selects verified courses within the user's budget and weekly time, with each step linked to UN SDG targets.

HR teams and career centres receive team skills matrices, cohort learning plans and progress tracking.

Responsible AI: the LLM performs only extraction and natural-language explanation; rankings, courses and costs are computed in code. Any model output containing figures absent from source data is rejected and replaced by a template.

Workshop skills applied: GenAI APIs (DeepSeek API via an OpenAI-compatible interface), database architecture (PostgreSQL schema for accounts, profiles, progress and time-series trend snapshots via SQLAlchemy) and cloud hosting (Replit Autoscale Deployments, Secrets management and scheduled data refresh).

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Frontend: HTML, CSS and vanilla JavaScript; three languages (English, Bahasa Malaysia, Chinese); works on phones.

Backend: Python, FastAPI, Uvicorn; deterministic scoring and pathway engine.

Database: PostgreSQL on Replit (SQLite locally) through SQLAlchemy.

AI: DeepSeek API (deepseek-chat model, OpenAI-compatible); for local runs, the WorkBuddy (Tencent CodeBuddy) CLI with the DeepSeek V4.1 Flash model can be used instead of an API key; templates when no model is available.

Data and APIs: ESCO v1.2 API, ILO Working Paper 140 GenAI exposure scores, UN SDG API, Hacker News (Algolia) “Who is hiring?” threads, arXiv API; optional Adzuna job ads.

Hosting: Replit Autoscale deployment with Secrets; daily trend refresh. During development, ngrok was used only to share a temporary public link to a locally running copy for testing.

System block diagram:

Browser -> Replit: FastAPI app -> PostgreSQL (accounts, profiles, trend snapshots)

FastAPI app -> DeepSeek API (reads CVs, rewords explanations)

FastAPI app -> ESCO API (live job search)

Daily refresh job: Hacker News + arXiv -> dated trend snapshots in PostgreSQL

Sponsor resources: Yes. Replit hosts the live app, its PostgreSQL database and Secrets. The language model comes from the DeepSeek API, which is not a sponsor resource.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Empathise: we wrote three synthetic personas from the brief's user groups: a mid-career office clerk, a fresh graduate switching fields and a TVET student, plus HR and career-centre users.

Define: each needs to know which skills are fading, which role fits, and what to do first, with reasons they can trust.

Ideate and prototype: we built the full chain end to end early, then improved it screen by screen: plain-language "Simple" mode, a four-step onboarding story, a guided tour, and team views.

Test: 64 automated tests cover scoring, fairness (age band never changes matches), model guardrails and data checks; we reviewed every screen in light, dark and phone layouts and fixed every problem we found.

Deployment: the prototype already runs on Replit with PostgreSQL and Secrets. Trend data refreshes daily, and only public or synthetic data is used. Phase 2 work: a Malaysian job-ad source, member sign-in for teams, a consent and deletion flow under the PDPA, and a pilot with a career centre.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4: "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Also Target 8.5: "full and productive employment and decent work for all".
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	More than 600,000 workers in Malaysia are expected to need reskilling within 3-5 years (MOSTI, January 2025, reported by Malay Mail). Signals are already visible: the share of job posts asking for spreadsheets and data handling fell from 2.3% to 1.6% in three months (Hacker News "Who is hiring?", May-Jul vs Aug-Oct 2026).

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	The % of users who leave onboarding with a ranked career and a costed reskilling path, and % who complete their first path step within 30 days.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	In a Phase 2 pilot with a career centre: 80% of pilot users leave onboarding with a ranked career and a costed path, and 30% complete their first path step within 30 days.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

6.3 Employment-Related IP Conflicts

Name	Type (code/dataset/model)	License
FastAPI 0.148.0	Code	MIT
Uvicorn 0.54.0	Code	BSD-3-Clause
HTTPX 0.28.1	Code	BSD-3-Clause
SQLAlchemy 2.1.5	Code	MIT
psycopg 3.3.6	Code	LGPL-3.0
RapidFuzz 3.14.6	Code	MIT
pypdf 6.21.1	Code	BSD-3-Clause
python-docx 1.2.0	Code	MIT
python-multipart 0.0.32	Code	Apache 2.0
openpyxl 3.1.5 (seed-data build tool)	Code	MIT

pytest 9.1.1 (testing)	Code	MIT
fpdf2 2.8.9 (sample CV build tool)	Code	LGPL-3.0
GSAP 3.12.5 with ScrollTrigger	Code	GSAP Standard License
Lenis 1.1.20	Code	MIT
Newsreader and Figtree fonts (Google Fonts)	Code (font)	SIL Open Font License 1.1
ESCO v1.2 occupations and skills (European Commission)	Dataset	EU reuse policy (Decision 2011/833/EU), with attribution
ILO Working Paper 140 GenAI exposure scores (Gmyrek et al., 2025)	Dataset	CC BY 4.0
UN SDG goals and targets (UN Statistics Division SDG API)	Dataset	UN terms of use
Hacker News “Who is hiring?” posts via Algolia HN Search API (counts only)	Dataset	Public API terms
arXiv API paper metadata (counts only)	Dataset	CC0 (metadata)
Adzuna API job-ad counts (optional, off by default)	Dataset	Adzuna API terms
Course and certificate catalogue (71 items, links to providers’ public pages)	Dataset	Each provider’s terms; names and links only
Landing-page background image by Richard Horvath (Unsplash)	Dataset (image)	Unsplash License
DeepSeek chat model (deepseek-chat, via DeepSeek API)	Model	DeepSeek API terms of service
WorkBuddy / CodeBuddy CLI (Tencent) with DeepSeek V4.1 Flash, local runs only	Model	Tencent CodeBuddy terms of service

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



Date: 24 October 2026